

Tejas Satish Navalkhe

Hyderabad, Telangana, India

☎ +91 7264 800601 ✉ tejas@tejasnavalkhe.com 🌐 @tejasnavalkhe 📧 @tejasnavalkhe 🌐 www.tejasnavalkhe.com

Professional Summary

AI/ML Engineer with 3+ years of experience developing Machine Learning, Predictive Analytics, Statistical Modeling, and AI-driven automation solutions across Banking, FinTech, and Transportation. Experienced in AML, KYC, Transaction Monitoring, Risk Analytics, Regulatory Reporting, and Python/SQL-based analytics. Skilled in developing scalable ML models, data analysis, and banking compliance solutions using Python, SQL, TensorFlow, PyTorch, and MLOps.

Work Experience

Deputy Manager (Technology)

August 2025 – Present

Jalgaon Janata Sahakari Bank Ltd.

Jalgaon, Maharashtra, India

- Automated **banking analytics**, MIS, RBI regulatory reporting, and **data analysis** across 10+ departments using Python, SQL, FastAPI, Pandas, and Matplotlib, reducing manual effort by 70% while improving **reporting accuracy by 90%**.
- Developed **AML analytics**, **transaction monitoring**, and **fraud detection support workflows** by automating Internal Account Monitoring, Enhanced Due Diligence (EDD), Suspicious Transaction Reports (STR), and FIU-IND reporting, reducing preparation time from 3 days to under 10 minutes while improving regulatory compliance.
- Designed secure on-premises automation infrastructure supporting **banking data pipelines**, regulatory reporting, compliance analytics, and RBI cybersecurity guidelines.
- Built internal banking automation platforms including Complaint Tracking, **Transaction Monitoring**, ATM/UPI Technical Decline Reporting, and operational dashboards supporting risk monitoring and digital banking operations.

Data Science Intern

May 2024 – September 2024

Co Wheels Car Club

Newcastle Upon Tyne, United Kingdom

- Designed **predictive machine learning models** using regression and ensemble techniques for dynamic pricing, analysing 30,000+ users and 600+ vehicles, increasing revenue by 19.32%.
- Performed feature engineering, statistical analysis, and automated preprocessing of 40+ features, improving model accuracy while reducing manual effort by 40%.
- Synced data with Co Wheels' booking system, reducing data sync issues by **25%**. Insights guided their shift to new pricing tool, boosting profits without compromising user satisfaction.

Freelance Software Engineer

September 2022 – April 2024

Finvestor Capital

Remote, India

- Architected an **algorithmic trading system** enabling **sub-millisecond execution**, increasing trading volume by **20%** and returns by **10%** in 3 months. Deployed on AWS/Docker for **50% faster scaling**, with Redis caching speeding data retrieval by **35%** and **99.9% uptime**.
- Built predictive ML models for risk analytics, algorithmic trading, and financial forecasting using Python, SQL, Redis, and AWS.
- Used **Redis pub/sub** to streamline multi-server data flow and refined SQL databases, cutting prediction latency by **20%** during high-frequency trading.

Web Development Intern

July 2022 – September 2022

Vitaecura Consultancy Services Pvt. Ltd.

Indore, Madhya Pradesh, India

- Revamped **5+ client-facing web features** (JavaScript, jQuery), improving page load by **20%** and engagement by **15%** via responsive design.
- Streamlined IT workflows to save **10 hours/month** in manual effort while supporting delivery of **3+ IT projects** using Agile.
- Drove business development** by creating client demo dashboards that secured **2 new clients**, and led UX research, improving site navigation to boost user satisfaction by **10%**.

Education

MS in Data Science (With Specialisation in Artificial Intelligence)

September 2023 – November 2024

Newcastle University, Newcastle Upon Tyne, United Kingdom

Grade: Distinction

- Modules:** Machine Learning, Deep Learning, Big Data Technology, Statistics, EDA, Data Management.

B. Tech in Computer Science Engineering

August 2019 – June 2023

Prestige Institute Of Engineering, Management & Research, Indore, India

CGPA: (8.78/10)

- Modules:** Data Structure, Cloud Computing, Project Management, Machine Learning, DBMS, Software Engineering.

Projects

Scalable AI-Driven Algorithmic Trading System with Real-Time Automatic Trading

- Constructed a low-latency algorithmic trading system using Apache Spark Streaming and WebSockets, integrating **Time Series Forecasting**, **Predictive Analytics**, and Statistical Modeling (LSTM, ARIMA) with **real-time sentiment analysis** to improve trade accuracy by 15% while reducing data latency by 40%.
- Containerised deployment with **Docker on AWS**, achieving **99.9% uptime**, and designed a custom risk-management module to limit losses to **¡2% per trade** while optimizing returns.
- Reverse-engineered **AngelOne/Zerodha APIs** to unify order execution across brokers and implemented a Selenium-based testing framework, cutting execution bugs by **30%** and ensuring compliance with trading regulations.

AI-Driven Face Recognition Attendance System

- Developed a **face recognition-based attendance system** using **OpenCV**, **automating attendance tracking** and reducing manual effort by **70%**.
- Achieved an impressive **99.38% recognition accuracy**, ensuring **reliable and consistent identification** under varied real-world conditions.
- Integrated **Amazon S3** for **large-scale image storage**, enabling **scalable, cloud-ready deployment** with **fast and efficient data access** for real-time face matching.

Skills

Programming Languages: Python, SQL, JavaScript

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, PyTorch, Regression, Classification, Statistical Modeling, Feature Engineering, Model Evaluation, Predictive Modeling, XGBoost, LightGBM, CatBoost

Banking & Risk Analytics : AML, KYC, Fraud Detection, Fraud Analytics, Transaction Monitoring, Risk Analytics, Predictive Analytics, Banking Compliance, RBI Reporting, FIU Reporting, EDD, STR, Banking Data Analysis

Natural Language Processing: Transformers, Hugging Face, spaCy, OpenAI APIs, BERT, GPT

MLOps: MLflow, Weights & Biases (W&B), DVC, Airflow, Docker, Kubernetes

Cloud & DevOps: AWS, Docker, CI/CD (GitHub Action), Monitoring & Logging

Software Development & Testing: Clean Code Principles, Design Patterns, Unit Testing (Pytest), Integration Testing

Version Control & Collaboration: Git, GitHub

Data & Databases: MySQL, PostgreSQL, MongoDB, Redis

Web Technologies & API: RESTful API, WebSocket, FastAPI, Flask

Data Engineering & Processing: Pandas, NumPy, Apache Spark, ETL Pipelines

Visualisation: Matplotlib, Seaborn, Plotly, Power BI

Awards & Achievements

NCL+ Award

Newcastle University

May 2024

Newcastle Upon Tyne, United Kingdom

Best Student Of the Batch (2019-2023) Award

Prestige Institute Of Engineering, Management & Research

April 2023

Indore, India

Key Certifications

- **International Conference on Cognition, Human and Artificial Intelligence** by KBCNMU — January 2026
- **Smart Banking with AI – A Deep Dive for Bankers** by Indian Institute of Banking & Finance — September 2025
- **Machine Learning with Scikit-Learn** by Madecraft — August 2024
- **Data Science & Advance Machine Learning using Python** by Grras Solutions — January 2022

Publications

- “**Design and Development of Smart Virtual Assistant Using Latest Tools and Technologies**”, International Journal of Core Engineering & Management, Feb. 2022, ISSN: 2348-9510